

Weighted average Cost of Capital

Methods of Computing Cost of Capital:

1. Cost of equity Capital.
2. Cost of debt Capital.
3. Cost of internally generated funds.
4. Overall Cost of Capital.
5. Cost of Preferred Capital.

Cost of equity Capital:

It is the cost of valuation and dividend it is estimated through equity sources.

$$K_e = \frac{E_a}{P_o}$$

Where K_e = Cost of equity Capital.

E_a = Expected avg earning per share.

P_o = Price of the share of equity if sold.

The expected avg earning per share of a Company is ₹ 8. & Current market price of the share is ₹ 80. Find out Cost of equity Capital.

$$K_e = \frac{8}{80} \times 100$$

$$= 10\%$$

A firm Current earnings are ₹ 40000 distributable among 4000 shares. The market price of each share is ₹ 125. The growth rate of dividend is 8%.

$$K_e = \frac{40000}{4000}$$

$$= 10$$

$$\text{Cost of Capital} = \frac{10}{125} + 8\%$$

$$= 0.16 \times 100$$

$$= 16\%$$

3. Current Earnings are ₹ 250000 to be distributed among 16000 shares the market price of each share ₹ 200 and Growth rate is 7%. Compute Cost of Eq C

$$K_e = \frac{250000}{16000} \quad C/P = \frac{15.625}{200} + 0.07$$

$$= 15.625 \quad 0.148 \times 100$$

$$= 14.8\%$$

4. A Company Current earnings are ₹ 1,25,000 to be distributed among 8000 shares. The market price of each share is ₹ 150 & Growth rate of dividend is estimated at 9%.

$$K_e = \frac{125000}{8000} \quad C/P = \frac{15.625}{150} + 0.09$$

$$= 15.625 \quad = 0.194$$

$$= 19.4\%$$

5. The market price of a share is ₹ 140, and a Company plan to pay dividend ₹ 9 per share. The growth in dividend is estimated 10%. Find out Cost of Equity Capital.

$$K_e = \frac{D_0}{P_0} + 0.10$$

$$= \frac{9}{140} + 0.10$$

$$= 0.064 \times 100$$

$$= 6.4\%$$

$$= 0.164$$

$$= 16.4\%$$

Cost of debt Capital :

$$K_d = \frac{\left(C + \frac{P-M}{N} \right) (1 - TR)}{\frac{(M+P)}{2}}$$

Where, K_d = Cost of Debt Capital

C = Annual interest

P = Par value. (face value)

M = Price of debenture.

N = No of years.

TR = Tax Rate.

20 years 12.5% Debentures of a firm are sold at a rate of 75. The face value of each debentures is 100. & TR is 50%.
Compute Cost of debt Capital.

$$K_d = \frac{\left(12.5 + \frac{100 - 75}{20} \right) (1 - 0.5)}{\frac{(75 + 100)}{2}}$$

$$= 0.0785$$

The Company has 8 years 15% Debentures of a firm sold at the rate of 80. The face value of debentures is 150. 50% Tax is assume. Find out Cost of debt.

$$K_d = \frac{\left(15 + \frac{150 - 80}{8} \right) (1 - 0.5)}{\frac{80 + 150}{2}}$$

$$= 0.1032$$

3. Cost of Internally generated fund: / Retain earning

Cost of retain earning is many firms that they new resort to a bank for financing of a new opportunity.

$$K_{re} = \frac{E_a \left(1 - \frac{TR}{2}\right)}{P}$$

K_{re} - Cost of retained earnings

E_a - Anticipated earnings

P - Market Price

TR - Rat of tax

1. The Current Market price of a share is ₹ 85. The Company anticipated earnings of ₹ 100000 to be distributed among 10000 share holders. The share holders tax rate is 30%. Find out Cost of retain earning.

$$\begin{aligned} K_{re} &= \frac{E_a \left(1 - \frac{TR}{2}\right)}{P} & E &= \frac{100000}{10000} \\ &= \frac{.10 \left(1 - \frac{0.30}{2}\right)}{85} & &= 10 \\ &= \underline{\underline{0.1}} \end{aligned}$$

2. The Current market price of the share is ₹ 125. A company earnings ₹ 250000 to be distributed among 15000 share holders tax rate is 40% you are required to find out Cost of retain earning.

$$\begin{aligned} E &= \frac{250000}{15000} & K_{re} &= \frac{E_a \left(1 - \frac{TR}{2}\right)}{P} \\ &= 16.67 & &= \frac{16.7 \left(1 - \frac{0.4}{2}\right)}{125} \\ & & &= \underline{\underline{0.106}} \end{aligned}$$

Overall / weighted average cost of Capital:

The Company has following Capital structure.

Securities	Book value (£)	After tax cost. (in%)
Equity Capital	850000	15.
Retain earnings	225000	10.
Preferred Capital	150000	18.
Debt	1000000	6.
	<u>2225000</u>	

From the above information find out WACC of Capital:

Securities	Book value	Proportionate	ATR	WACC
Eq. Cap	850000	38	15	5.7
Retain earnings	225000	10	10	1.0
Preferred Capital	150000	6.7	18	1.206
Debt	1000000	45	6	2.7
				<u>10.606</u>

The Company has following Capital structure find out WACC.

Securities	BV (£)	ATR (in%)
Eq. Cap	500000	13%
Retain earnings	200000	8%
Pay. Capital	200000	14%
Debt	400000	5%
	<u>1300000</u>	

Security	BV	Propor	ATC	WACC
Eq. Cap	500000	38	13	4.94
Ret. Earnings	200000	15	8	1.2
Pay. - Cap	200000	15	14	2.1
Debt	400000	31	5	1.55
	<u>1300000</u>			<u>9.79</u>

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Compute WACC Based on 1) BV 2) MV.			
Source of finance	BV	MV	ATC (%)
Eq. Cap	1200000	1300000	12
Retain long term Debt cap	900000	800000	8
Short term debt	250000	260000	4
	2350000	2360000	

Book value %				
Source of finance	BV	Proportion	ATC	WACC
Eq. Cap	1200000	51	12	6.12
LTD	900000	38	8	3.04
STD	250000	11	4	0.44
	2350000			9.6

Market value %				
Source of finance	MV	Proportion	ATC	WACC
Eq. Cap	1300000	55	12	6.6
LTD	800000	34	8	2.72
STD	260000	11	4	0.44
	2360000			9.76

13/09/23
4 ABC Company Supplied the following information

Compute WACC Based on BV & MV.

Source of finance	BV	MV	ATC (%)
Eq. Capital	1200000	1300000	5
Debt	400000	380000	8
Pref. sh.	100000	110000	15
Eq. sh.	600000	1200000	13
Retain earnings	200000	-	
Total	1300000	1690000	

Determine WACC using BV weight rate & MV rate.

Book value:

Source of finance	BV	Proportionate	ATC	WACC
Debt	400000	31	5	1.55
Pay sh	100000	8	8	0.64
Eq sh	600000	46	15	6.9
retain earning	200000	15	13	1.95
	1300000			11.04

Market value:

Source of finance	MV	Proportionate	ATC	WACC
Debt	380000	22	5	1.1
Pay sh	110000	7	8	0.56
Eq sh	1200000	71	15	10.65
retain earning	—			
	1690000			12.31

The NPCC as been following Capital structure

Particulars	Amnt.
Eq. Cap (20000 shares)	4000000
10% Pay Cap	1000000
14% Debentures	3000000
	8000000

The shares of the Company sells for ₹ 20 if it is expected that the Company will pay next year dividend of ₹ 2/sh which growth rate 7%. Assume 50% tax rate.

Compute WACC. Based on existing Capital structure.

Compute new WACC if Company raises additional ₹ 2000000 debt by issuing 50% debentures. This would increase the expected dividend of ₹ 3/sh and the growth rate and unchanged. But the Price of share will follow P/E ratio.

1) Cost of equity = $\frac{E}{P} + G$

$= \frac{21}{20} + 7\%$

$= 0.17$

Cost of Debt = $(14\% - 50\%)(14\%)$
 $= (0.14 - 0.07)$
 $= 0.07$

WACC	Proportionate	Cost %	WACC
40 000 000	50	0.17	8.5
10 000 000	13	0.10	1.3
30 000 000	37	0.07	2.59

12.39

2) Cost of Equity = $\frac{E}{P} + G$

$= \frac{3}{15} + 0.07$

$= 0.27$

Cost of Debt = $150\% - 50\% = 150\%$
 $= 0.15 - 0.25 = 0.075$

WACC	Proportionate	ATC	WACC
40 000 000	40	0.27	10.8
10 000 000	10	0.10	1
30 000 000	30	0.075	2.1
20 000 000	20	0.075	1.5
100 000 000			15.4

Rajeev Company has been following Capital structure

5000 Eq. sh. ₹ 100 each 500000.

15% Paj sh ₹ 300000.

10% Debentures ₹ 600000.

The Current market price is ₹ 95. a Company is expected to declare a dividend of ₹ 15. at the end of Current year with an expected growth rate of 12%.

The tax rate is applicable 40%. Calculate Cost of Eq & WACC.

assuming the Company can raise ₹ 400000. 10% Debent.

Find out new WACC. If dividend rate increases by ₹ 18 and growth rate is reduce from 5% & market Price reduced to ₹ 80.

$$\text{Cost of equity} = \frac{E}{P} + G$$

$$= \frac{15}{95} + 0.12$$

$$= 0.277$$

$$\text{Cost of Debt} = (10\% - 40\% \text{ of } 10\%)$$

$$= 0.1 - 0.04$$

$$= 0.06$$

			Proportionate	WACC
Eq. sh	500000	0.277	36	9.972
Paj sh	300000	0.15	21	3.15
Debentures	600000	0.06	43	2.58
	1400000			15.702

$$\text{Cost of equity} = \frac{E}{P} + G$$

$$= \frac{18}{80} + 0.05$$

$$= 0.275$$

$$\text{Cost of Debt} = (10\% - 40\% \text{ of } 10\%)$$

$$= 0.1 - 0.04$$

$$= 0.06$$

Eq.sh	500000	28	0.275	7.7
Pr.sh	350000	16	0.15	2.4
Debtors	600000	33	0.66	21.78
New Debentures	400000	22	0.66	14.52
	1800000			<u>46.4</u>